



INSTITUTE OF SPACE TECHNOLOGY
KICSIT, Kahuta Campus



PROJECT PROPOSAL

Scientific Evaluation Framework for Benchmarking Free Large Language Models

Prepared by:

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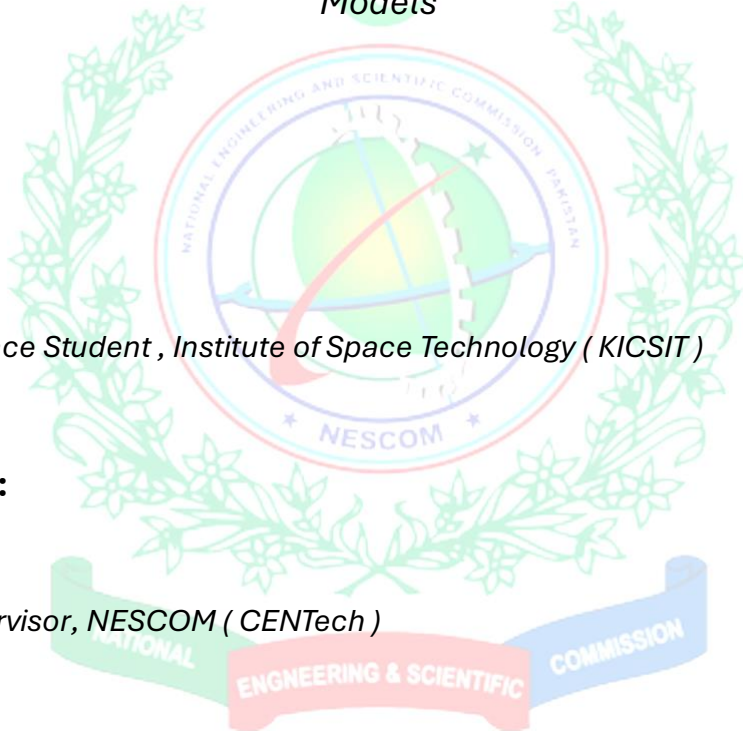
Submitted to:

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Date:

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1. Project Goal

To develop a reproducible benchmarking framework that scientifically compares freely accessible LLMs across multiple dimensions. The framework will be built upon comprehensive, custom-developed benchmarking techniques to ensure objective performance evaluation.

2. Scientific Methodology

To ensure scientific rigor, the project will follow this structured approach:

- **Research Questions:** Evaluate models based on accuracy, reasoning, hallucination rates, instruction following, robustness, long-context retention, stability, and speed vs. quality.
- **Variables:** Define specific independent variables (e.g., model, temperature, prompt complexity) and dependent variables (e.g., latency, accuracy, consistency).
- **Controlled Environment:** Document and hold constant hardware, OS, Python/SDK versions, and network configuration to ensure full reproducibility.
- **Model Verification:** Verify model IDs via metadata/headers and capture visual documentation (screenshots) of model selection to account for silent provider updates.

3. Dataset & Benchmark Categories

We will construct an original, balanced dataset of **220 prompts** to mitigate benchmark contamination. To ensure fair and statistically significant scoring, every subset contains at least 20 prompts:

- **Core Assessment:** 30 Knowledge Retrieval, 30 Multi-step Reasoning, 30 Instruction Following.
- **Specialized Stress Tests:** 20 each of Ambiguity Handling, Hallucination Stress Tests, Long-context Retention, Coding, Data Transformation, Summarization, and Multilingual tasks.

4. Evaluation & Statistical Analysis

- **Scoring:** A 0–5 rubric will measure factual correctness, completeness, relevance, instruction compliance, reasoning quality, clarity, formatting, safety, hallucination avoidance, and conciseness.
- **Reliability:** Prompts with stochasticity will be repeated 5–10 times to compute statistical measures (mean, median, standard deviation, coefficient of variation).
- **Analysis:** Perform confidence interval calculations, hypothesis testing, correlation, regression, and inter-rater agreement analysis.
- **Human-in-the-Loop:** In Week 6, we will compare automated scores against human-evaluated scores to validate our framework’s accuracy.

5. Feasibility & Operational Constraints

- **Free-Tier Limitations:** This project relies on free-tier LLM access. To address potential rate limits or service quotas, the framework will utilize an **iterative, batch-based testing system** with intelligent throttling and exponential backoff to ensure successful data collection without bottlenecking.
- **Approval Requirement:** We are seeking formal approval for this strategy to ensure the project timeline remains viable under service usage constraints.

6. Weekly Execution Timeline

Phase	Duration	Focus Area
Design	Week 1	Hypotheses and control variables.
Framework	Week 2	Automation scripts and logging systems.
Dataset	Week 3	Curation of 220 original prompts.
Experiments	Weeks 4 - 5	Execution and raw data collection.
Human Eval	Week 6	Manual calibration of automated scores.

Phase	Duration	Focus Area
Statistics	Week 7	Formal statistical analysis.
Reporting	Week 8	Final comprehensive report (50-80 pages) and presentation.

7. Deliverables

- **Research Report:** A comprehensive 50–80 page document covering methodology, experimental results, statistical analysis, and ethical implications.
- **Reproducibility Package:** Complete benchmark framework, original dataset, automation scripts, results repository, and final presentation.
- **Weekly Reporting:** Includes screenshots of experiments and detailed progress updates for the supervisor.

